

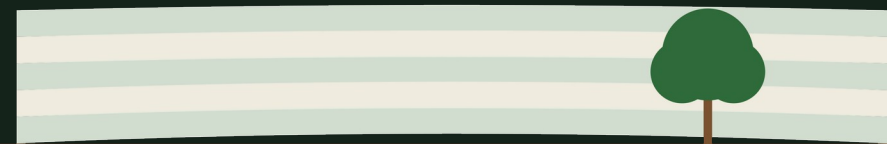
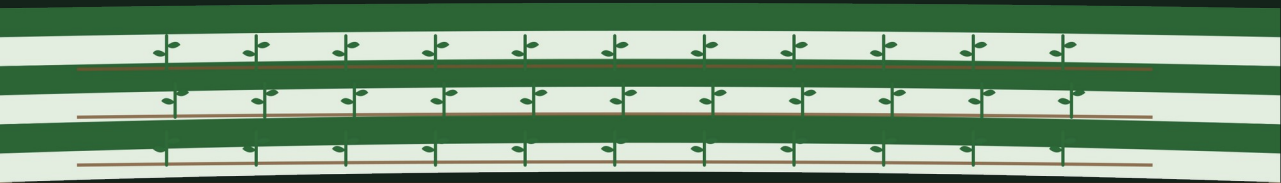


DEALER TRAINING · SEPTEMBER 2026

Humi Grow Plus

Liquid potassium humate, humic acid 10% and K_2O 3.5%, 4 L, for the irrigation line at germination, made in Lahore.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

Humi Grow Plus is potassium humate as a liquid: humic acid 10% and K₂O 3.5%, in a 4 L pack. It is the same active as the Humi Grow pellet, in the form that goes down an irrigation line. Humic substances add exchange sites to a soil, so potash, calcium, magnesium and ammonium are held for the root instead of leaching, and they chelate zinc, iron, manganese and copper, so they stay in solution at pH 8. Across 81 published studies, humic raised root and shoot growth. The liquid is put on at germination and early growth, while the root system is being built, and it lands in the wetted zone the roots occupy, across the whole acre. Wheat is 1 pack; maize and potato are 2. VAN has made humic acid here since 2007. It is a soil and root benefit. It does not shield phosphate from calcium, it will not move a soil test, and VAN prints no percentage saved.

The 4 things it must contain

1

The pack: humic acid 10%, K₂O 3.5%, 4 L, liquid, PS 5610-2023. Dilutes and goes down the line; no dissolving step.

2

The mechanism: exchange sites that hold nutrient, chelation that keeps micronutrients in solution, and root growth. A soil and root benefit, placed where the roots are.

3

The stage: germination and early growth, through the water, 1 to 2 packs an acre. The root that meets it is the one being built.

4

The limit: not a calcium shield, no percentage saved, no soil-test change. The pellet form is Humi Grow, at land preparation.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the pack says, and what it is registered as.

4 L

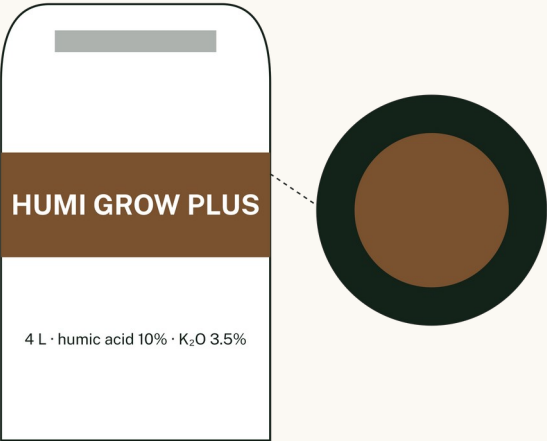
the pack. About 400 g of humic acid and 140 g of K₂O in it, reading the label percentages as weight per volume.

10%

humic acid, as the potassium salt, in solution. Dilute, and delivered in the water where the roots are.

K₂O 3.5%

the potash that comes with the humate. Counted on the programme; never the potash line.



Liquid

pours into the drum or the injector and goes down the line. No dissolving, no spreader pass.

2007

the year VAN's founder made humic acid from Pakistani leonardite. By 2012 VAN was the largest producer in Pakistan. It makes potassium humate in crystal, powder, pellet and liquid

0.61%

mean organic matter across 770,160 Punjab samples; 97.6% under 1%. The fraction this pack carries is the one most fields have lost.

Registered as	Pakistan Standard	Manufacturing licence
Humi Grow Plus	PS 5610-2023	CM/L-3274/2024

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

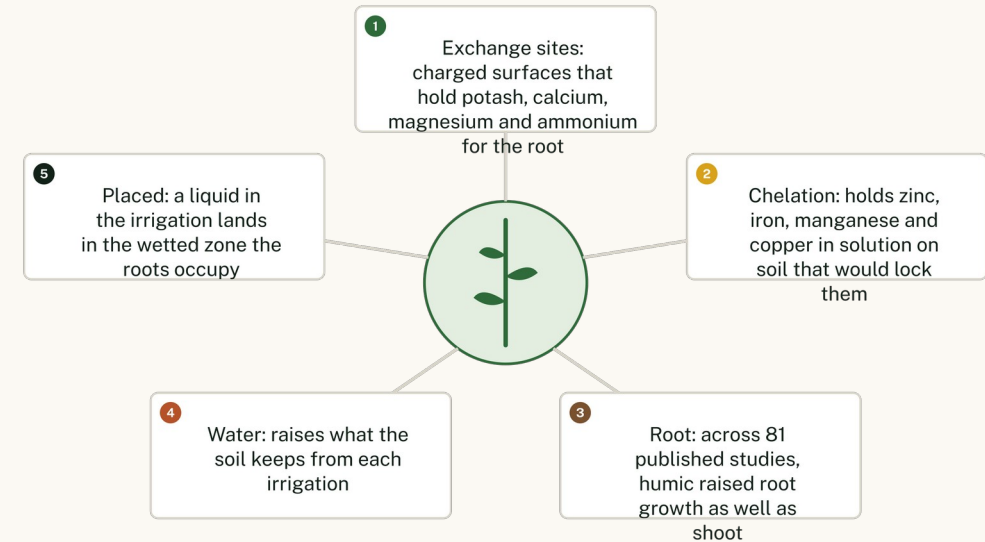
“Everything on my potato goes through the drip now. Is there anything for the soil itself I can put through the line, or is the soil only for the spreader?”

He has moved his nutrition into the water and his soil amendments have not followed, because the pellet needs a drill or a spreader and a ploughing. The answer is 3 ideas: what a humate does in a soil, why the liquid goes in with the crop already standing, and what it will not do.

The 3 answers on the next 3 slides, 1 idea each.

A humate gives the soil somewhere to hold nutrient, and keeps micronutrients in solution.

Humic substances are the long-lived fraction of organic matter: large molecules covered in charged sites. Those sites hold potash, calcium, magnesium and ammonium where the root can reach them. The same molecules wrap around zinc, iron, manganese and copper, which chemists call chelation, and hold them in solution on ground that would otherwise precipitate them at pH 8. And across 81 published studies, humic raised root growth as well as shoot growth. That is what the site prints for VAN's humates: applied nutrients held on the exchange sites instead of leaching or precipitating, and a root benefit.



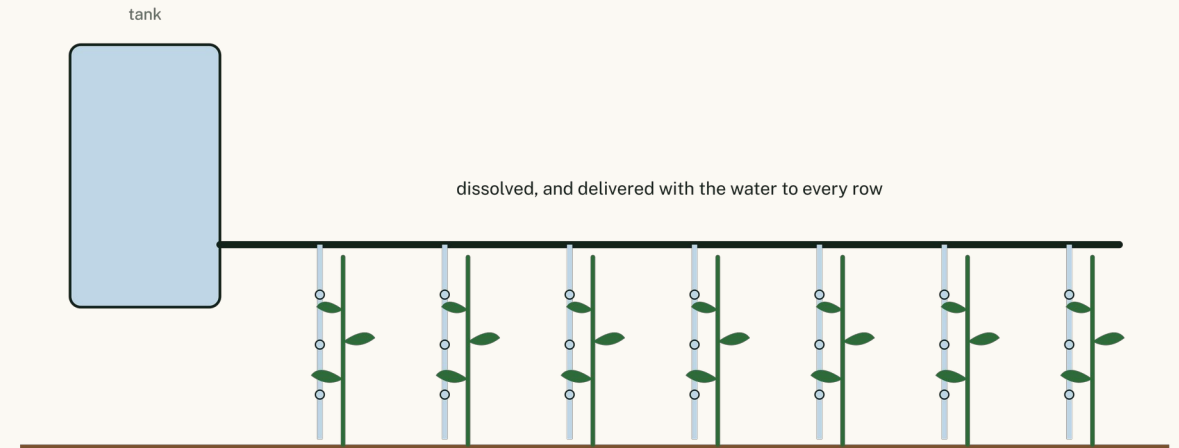
The 5 jobs of a humate, textbook chemistry and the 81-study finding. A picture of the idea, not a measurement of this product.

Say it like this: “Your soil has no hands to hold what you give it. A humate is hands, and this is the form that goes down your drip line.”

A liquid lands in the wetted zone, with the crop already in.

The pellet goes on before the crop, with the drill or the spreader, and is worked into the top few inches. A liquid injected with the irrigation goes where the water goes: into the wetted zone the roots occupy, across the whole acre, with the crop already standing. So the programme puts Humi Grow Plus on at germination and early growth, when the root system is being built and the soil around it decides what that root meets. Wheat, garlic and citrus take 1 pack at germination; maize takes 2 at germination; potato takes 2 at early growth; strawberry takes 2 at early growth and 2 more at maturity.

Say it like this: “It goes in at the drum, with the water flowing, when the crop is young and the roots are being made.”

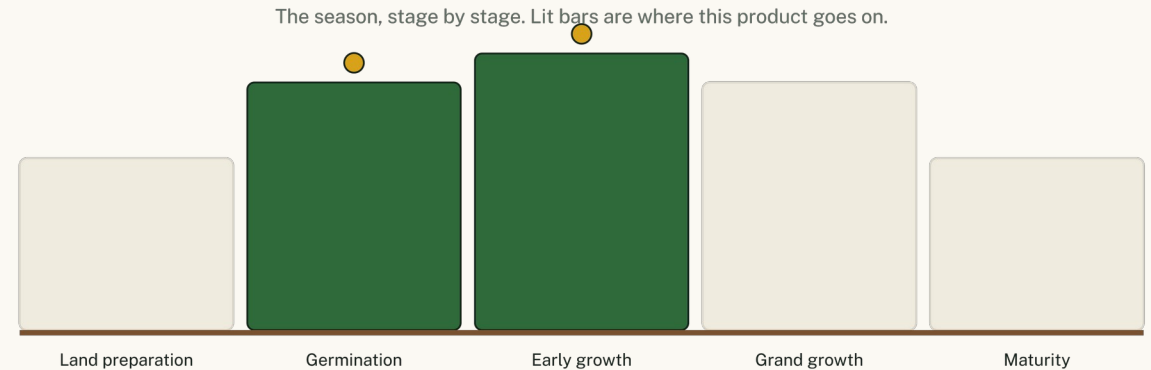


The fertigation line, from the drum at the head of the watercourse to the wetted root zone. Which stage carries it for your crop is on the programme slides.

What it will not do: shield phosphate from calcium, or move a soil test.

A humate is often sold as the thing that keeps calcium away from phosphate. VAN checked that against its own soil numbers and it fails by orders of magnitude: at any rate a grower would apply, humic acid binds too little calcium to matter to the phosphate. So VAN does not say it, and prints no percentage of phosphate saved. 1 pack is about 400 g of humic acid on an acre, so the organic-matter reading on the report will not move either. What changes is the wetted root zone the young crop grows into. The 3.5% K₂O is 140 g a pack: counted, and never the potash line.

Say it like this: “It helps the root and the soil around it. It does not stop your DAP being fixed, and it will not change your soil test. VAN says both out loud.”



The stage bar: germination and early growth are the programme's windows. The limits are printed because a dealer who overclaims a humate loses the grower at the next soil test.

What is verified for Humi Grow Plus, and what is not.

No trial of Humi Grow Plus itself is on VAN’s record. What a dealer may state as fact is the pack, the licence, VAN’s own history with humic acid, the soil record and the published research below, each as what it is.

Item	What it says	How a dealer may use it
The pack	Humi Grow Plus, liquid potassium humate, humic acid 10%, K ₂ O 3.5%, 4 L.	As fact.
The licence	Pakistan Standard PS 5610-2023, manufacturing licence CM/L-3274/2024.	As fact. The number, never an expiry date.
VAN’s record	Humic acid made in Pakistan from Pakistani leonardite by VAN’s founder in 2007; VAN the largest producer of humic acid in Pakistan by 2012; potassium humate made today in crystal, powder, pellet and liquid.	As fact, from the company timeline on van.com.pk .
The soil record (Punjab, 770,160 samples)	Mean organic matter 0.61%; 97.6% under 1%. 59.0% low in zinc; mean pH 8.22.	As fact about the soil. Never as product performance.
Published research, not VAN trials	Rose et al. 2014, a meta-analysis of 81 studies: humic raised shoot growth by about 22% and root growth by about 21%.	Spoken, as humic research. Never as a Humi Grow Plus result.
VAN’s own check	VAN’s 2026 stoichiometry check: at field rates humic acid binds too little calcium to shield phosphate. VAN does not make the calcium-chelation claim.	As the reason the shield claim is never made.
Not on file	A Humi Grow Plus trial; a yield figure; a nutrient-use-efficiency percentage; a soil-test change; a percentage of phosphate or nitrogen saved; a foliar rate; a 20 L price.	Not said until named.

The pack, and the acre, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. Humic acid and K₂O from the declared analysis, read as weight per volume. Wheat is 1 pack; maize and potato are 2; strawberry is 4 across the season.

Humi Grow Plus, humic acid 10% · K ₂ O 3.5%	Volume	Humic acid	K ₂ O	Rupees
1 pack of 4 L	4 L	400 g	140 g	1,240
Wheat, garlic, citrus: 1 pack at germination	4 L	400 g	140 g	1,240
Maize at germination, potato at early growth: 2 packs	8 L	800 g	280 g	2,480
Strawberry: 2 packs at early growth, 2 at maturity	16 L	1.6 kg	560 g	4,960

On wheat the grower spends 1,240 rupees an acre to put about 400 g of humic acid into the wetted root zone at germination. On potato he spends 2,480 for 800 g at early growth. The K₂O is counted and is never the potash line. That is the counter arithmetic. What the soil and the root do with it is not a number VAN prints for this product.

Price: the VAN farmer price sheet, Aug 2026. 400 g = 4 L × 10% w/v; 140 g = 4 L × 3.5% w/v. Rates: VAN’s published crop programmes on van.com.pk.

The pellet and the liquid on the same programme: 2 forms, 2 stages, added together.

The wheat, maize, potato, garlic and citrus programmes carry both humate forms. Rupees at the VAN farmer price sheet, Aug 2026.

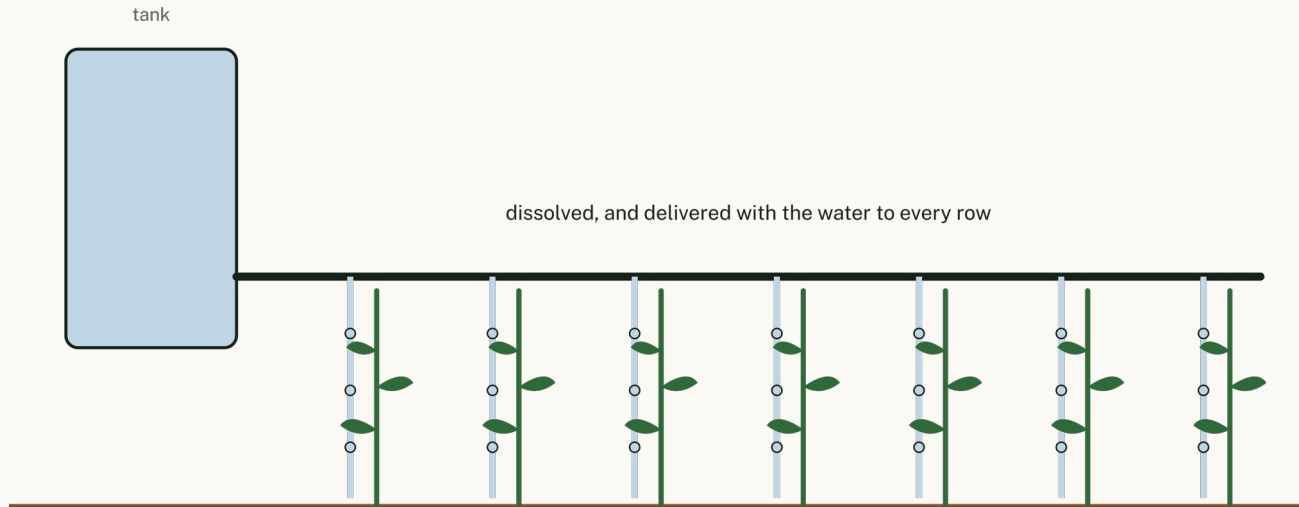
Crop	Humi Grow, 8 kg pellet	Humi Grow Plus, 4 L liquid	Stages	Rupees per acre, both
Wheat	1 bag, land preparation, drill at sowing (or broadcast)	1 pack, germination, fertigation	Before sowing; germination	3,990
Maize	1 bag, land preparation, drill at sowing (or broadcast)	2 packs, germination, fertigation	Before sowing; germination	5,230
Potato	1 bag, land preparation, drill at sowing (or broadcast)	2 packs, early growth, fertigation	Before sowing; early growth	5,230
Garlic	1 bag, land preparation, drill at sowing (or broadcast)	1 pack, germination, fertigation	Before sowing; germination	3,990
Citrus	2 bags, land preparation, drill at sowing (or broadcast)	1 pack, germination, fertigation	Before the season; germination	6,740

The programme adds the liquid to the pellet; it never swaps one for the other. The pellet is a bulk soil application before sowing; the liquid is placed in the wetted zone with the crop already in. Do not compare them on the percentage alone: different form, different rate, different stage.

Rows as printed in VAN's published crop programmes on van.com.pk. Humi Grow 8 kg PKR 2,750; Humi Grow Plus 4 L PKR 1,240 (the VAN farmer price sheet, Aug 2026). Wheat $2,750 + 1,240 = 3,990$; maize $2,750 + 2,480 = 5,230$; citrus $5,500 + 1,240 = 6,740$.

HOW IT GOES ON

Through the water, at germination and early growth. Into the drum, then across the whole irrigation.



Measure it, then dilute it

1 pack of 4 L at germination on wheat, garlic and citrus; 2 packs on maize at germination and on potato at early growth. Into a 200 L drum of water at the head of the watercourse, or the injector tank on a drip system, and stirred in. The drum is drawn at 200 L because that is what stands at the head of most watercourses; scale the water, never the product.

Meter it in, with the water already flowing

Open the tap so the drum empties across the whole irrigation, not in 1 slug. The liquid goes where the water goes: into the wetted root zone, across the whole acre. A slug at the start feeds the first strip and starves the last.

At the row your plan prints; never as a foliar from this deck

Germination and early growth are the windows on every field-crop row, while the root system is being built. Strawberry takes it again at maturity and the sugarcane ratoon at grand growth, as printed. Every row says fertigation. No foliar rate is printed, because no programme row names one.

“1,240 for 4 litres, and it is only 10%?”

The dealer says, out loud

Write the price and then write what it does: about 400 g of humic acid, in solution, put into the wetted root zone at germination, across the whole acre, with the crop already standing. 1 pack on wheat, 2 on potato. The pellet is 40% because it goes on dry before sowing and is worked in; the liquid is 10% because it goes down the line, placed where the roots are, when they are being built. Different form, different stage. What the soil and the root do with it is not a number VAN prints.

The fact behind it

- 4 L, 1,240 rupees, delivered (the farmer price sheet, Aug 2026).
- Wheat 1 pack; maize and potato 2 packs (the crop programmes on van.com.pk).
- About 400 g of humic acid and 140 g of K₂O per pack, reading the label as w/v.
- Humic acid made in Pakistan by VAN's founder since 2007 (the company timeline on van.com.pk).

Never say: “It pays for itself in saved DAP” or any rupees-per-acre return. No percentage saved and no yield figure exists for the product.

“If it unlocks my phosphate, can I cut a bag of DAP?”

The dealer says, out loud

No, and please do not tell a grower it unlocks his phosphate. VAN checked whether humic acid at field rates can keep calcium away from phosphate, and the answer was no by a wide margin, so VAN does not claim it and prints no percentage saved. What a humate does is hold nutrient on exchange sites, keep micronutrients in solution, and help the root. Keep the phosphate line as the programme prints it.

The fact behind it

- Humic is a soil and root benefit; the calcium-shield mechanism is not on van.com.pk.
- VAN's 2026 stoichiometry check: humic acid binds too little calcium at any field rate to shield phosphate.
- No percentage of phosphate saved is printed anywhere by VAN.

Never say: “It saves X% of your phosphate” or “cut the DAP by a bag”.

OBJECTION 3 OF 8

“I used it and my soil test did not change.”

The dealer says, out loud

It will not, and VAN says so. 1 pack is about 400 g of humic acid on an acre of soil, and a soil test measures the whole acre to plough depth. What changes is the wetted root zone the young crop grows into: exchange sites that hold what you apply, micronutrients kept in solution, and the root. Judge it there, and keep the soil test for its own sake.

The fact behind it

- About 400 g of humic acid per pack (the declared analysis, read as w/v).
- Mean organic matter 0.61%; 97.6% of samples under 1% (the Punjab soil record, 770,160 samples).
- A rise in extractable phosphorus after adding humic acid is not by itself proof of availability (general soil chemistry).

Never say: “Your organic matter will rise by X” or “the report will read higher next season”.

“Can I spray it on the leaf instead?”

The dealer says, out loud

Every programme row for Humi Grow Plus says fertigation, and the label names fertigation as the method, so that is what this deck prints. It is made to go into the soil solution where the roots are. If a grower wants a foliar product for the leaf, the programme names other products for that, at their own rows. Do not give a foliar rate for this one from this deck.

The fact behind it

- The method on the label and the product page is fertigation; all 8 programme rows print fertigation (van.com.pk).
- The foliar rows on the same programmes belong to the VL liquids and Tornado.
- No foliar rate for Humi Grow Plus is published anywhere.

Never say: A foliar rate, such as “1 L in 100 L of water”.

“Which stage, and how many packs?”

The dealer says, out loud

Germination on wheat, garlic and citrus, 1 pack; germination on maize, 2 packs; early growth on potato, 2 packs; strawberry 2 packs at early growth and 2 at maturity; the sugarcane ratoon 2 packs at grand growth. Through the water in every case, with the crop already in. Early, because the root that meets it is the one being built. Not at maturity on a field crop: that is late for the job it does.

The fact behind it

- Rates per acre as printed in the crop programmes on van.com.pk; 7 programmes on the programme slides.
- The product page gives the stage as early growth.
- The root system is built in the first weeks; a humate placed then meets the root being made.

Never say: “3 packs for a bad soil.” The programme’s rate is the rate; no dose response is on file.

“It has potash in it. Do I count it?”

The dealer says, out loud

Count it, and do not lean on it. K_2O 3.5% of 4 L is about 140 g a pack. The potash line on the programme is a separate bag at land preparation and another through the water at fill; this one adds a few grams to the account and changes neither. Write it down and move on.

The fact behind it

- About 140 g of K_2O per pack: $4\text{ L} \times 3.5\% \text{ w/v}$ (the declared analysis).
- The potash line on the same programmes is Fusion Potash or SOP at land preparation and Vital Potash through the water at grand growth (the crop programmes on van.com.pk).
- Too much potash pushes out magnesium and calcium; at 140 g this pack does not move that account.

Never say: “It is a potash fertilizer too” or that it replaces any part of the potash line.

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 5610-2023, manufacturing licence CM/L-3274/2024, VAN as the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the pack and VAN sends you that batch's certificate, free.

The fact behind it

- Humi Grow Plus: PS 5610-2023, CM/L-3274/2024 (the regulatory table on van.com.pk).
- Humi Grow, the pellet, carries the same standard under CM/L-3281/2024.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017.

Never say: An expiry date, or the licence as a claim about performance.

“I already put the Humi Grow pellet on at land preparation. Why the liquid too?”

The dealer says, out loud

Because the programme puts them at 2 stages for 2 reasons. The pellet went in before sowing, drilled or worked into the top few inches across the field. The liquid goes in at germination or early growth, with the crop standing, and lands in the wetted zone where the young roots are, which the drill and the spreader cannot reach once the crop is in. Wheat carries both: 1 bag of the pellet, then 1 pack of the liquid, 3,990 rupees an acre for the 2. Neither is swapped for the other.

The fact behind it

- Wheat, maize, potato, garlic and citrus carry Humi Grow at land preparation and Humi Grow Plus at germination or early growth (the crop programmes on van.com.pk).
- Humi Grow 8 kg 2,750 and Humi Grow Plus 4 L 1,240: 3,990 on wheat (the farmer price sheet, Aug 2026).
- Same active, 2 forms, 2 stages.

Never say: “Skip the pellet if you use the liquid”, or the reverse, or “the two together give X”.

WHERE IT SITS IN THE CROP PROGRAMME

Humi Grow Plus in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Germination	1 Pack	Fertigation	4 L
Maize (Corn)	Germination	2 Pack	Fertigation	4 L
Potato	Early Growth	2 Pack	Fertigation	4 L
Garlic	Germination	1 Pack	Fertigation	4 L
Citrus, 5 years and over	Germination	1 pack	Fertigation	4 L
Strawberry	Early Growth	2 Pack	Fertigation	4 L
Strawberry	Maturity	2 Pack	Fertigation	4 L
Ratoon Sugarcane	Grand Growth	2 Pack	Fertigation	4 L

Rows as published in van.com.pk's crop programmes, all 7 programmes that name Humi Grow Plus. Fertigation on every row; germination or early growth on every field crop; strawberry again at maturity and the ratoon at grand growth. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield figure, or a nutrient-use-efficiency percentage. No Humi Grow Plus trial exists; the 81-study finding is humic research, spoken, never a product result.

2 That it keeps calcium away from phosphate, or saves any percentage of phosphate or nitrogen. VAN's own check bars the mechanism and the site prints no percentage saved.

3 A soil organic-matter or soil-test change. About 400 g of humic acid on an acre is a root-zone input.

4 A foliar rate. Every programme row and the label say fertigation.

5 That it replaces the pellet, or the pellet replaces it, or that the 2 together give a measured result.

6 A 20 L price, or that it is the potash line. Only the 4 L pack is confirmed, and the K₂O is 140 g a pack.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for Humi Grow Plus shows

The declared analysis tested for the batch: humic acid and K₂O against 10% and 3.5%, with the batch number, the Pakistan Standard PS 5610-2023 and licence CM/L-3274/2024 on the pack. No certificate for a Humi Grow Plus batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate. Commercial humic products elsewhere can deviate widely from label, which is the reason to ask for the certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

A humate gives the soil hands to hold what you apply. This one goes down the water line.

Urdu first pass, unreviewed

ہیو مک زمین کو ہاتھ دیتا ہے کہ کھاد کو تھامے رکھے، اور یہ پانی کی لائن سے جاتا ہے

Put it in the drum at germination or early growth, with the water flowing.

Urdu first pass, unreviewed

اُگاؤ یا ابتدائی بڑھوتری پر ڈرم میں ڈالیں، پانی چلتا رہے

It helps the root and the soil. It does not change your soil test, and VAN says so.

Urdu first pass, unreviewed

یہ جھڑ اور زمین کی مدد کرتا ہے۔ یہ آپ کی مٹی کی رپورٹ نہیں بدلتا، اور VAN یہ جھل کر کہتا ہے

The pellet goes in before sowing; the liquid goes in with the crop standing.

Urdu first pass, unreviewed

دانہ بوائی سے پہلے، مائع کھڑی فصل کے ساتھ

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

Humi Grow Plus: the humate fraction, in the water, at germination and early growth. A soil and root benefit. Not a calcium shield, no percentage saved.

